

Contents

Literature Survey: ML for FIFA Match Prediction	1
Key Takeaways	1
1. Team-Level Approaches (Older Method)	1
2. Player-Level Approaches (Better Method)	1
3. World Cup–Specific Studies	1
4. Graph Neural Networks (GNN) — The New Trend	2
5. Gaps in Existing Research (Why This Project Matters)	2
6. Summary Table	2
References	3

Literature Survey: ML for FIFA Match Prediction

Key Takeaways

- Older models only use **team stats** (rankings, past results).
- Newer models add **player stats** (goals, passing, tackles) for better accuracy.
- **World Cup prediction is harder** than league prediction because national squads change every tournament.
- **Draws are the hardest** outcome to predict in almost every study.
- **Graph Neural Networks (GNNs)** are the newest trend — they model player-to-player interactions, but are rarely used for World Cup prediction yet.

1. Team-Level Approaches (Older Method)

These models treat a team as one unit, using only history and rankings.

- **Rodrigues & Dias** – Logistic Regression + Decision Trees, focused on betting accuracy.
- **Atta Mills et al.** – Random Forest + Neural Network on Premier League data.
- **Zeileis & Hornik** – Random Forest using rankings + player market value (2022 World Cup).
- **Liu (2024)** – Random Forest trained on match history from 1872–2023.

Limitation: Doesn't know *which players* are actually playing — just team reputation.

2. Player-Level Approaches (Better Method)

These models add individual player stats.

- **Wong et al.** – Used goals, assists, passing accuracy per player.
- **Saha et al.** – Deep Learning (DNN/ANN), 63.3% accuracy on 2018 World Cup.
- **2023 Women's World Cup study** – XGBoost using official technical/physical stats, ~58% accuracy.
- **Frontiers (2025)** – Neural Network (MLP) using 22 technical indicators, 86.7% accuracy (highest so far).

Limitation: Draws are still hard to predict correctly in all these studies.

3. World Cup–Specific Studies

The World Cup is different from league football: - Squads are rebuilt every tournament. - Has group stage + knockout stage.

Support Vector machine

Key studies: - **TGM Research** – **SVM** + Neural Network, used only rankings/history. - **Oxford Model (Bull)** – Math/probability based, no player data. - **Al-Bustami & Ghazal (main paper)** – Combines player + team data, uses 5 models together (Logistic Regression, Random Forest, XGBoost, AdaBoost, KNN) with majority voting. - Result: **59.38% accuracy** vs **56.25%** baseline. - Best improvement in **low-scoring matches** (52% vs 48%).

4. Graph Neural Networks (GNN) — The New Trend

Calculates player synergy

GNNs treat players as **connected nodes** instead of just numbers in a table. This captures passing patterns and teamwork.

- **Huang et al.** – GNN used in American football & esports (not soccer yet).
- **HIGFormer (2025)** – Graph Transformer for club soccer prediction.
- **Ball-player GCN (2025)** – Tracks player movement + ball using graphs (91%+ accuracy on tracking tasks).
- **SFU Thesis** – Argues most models ignore which exact players are on the field — GNNs can fix this.

Limitation: GNNs are mostly tested on club football or single matches — **not yet applied to full World Cup prediction.**

5. Gaps in Existing Research (Why This Project Matters)

Gap	Explanation
Roster changes	National teams change every World Cup — most models don't adjust well
Draw prediction	Every model struggles here
Player + team fusion	Very few models combine both properly
GNNs for tournaments	GNNs work well but haven't been applied to World Cup outcome prediction

6. Summary Table

Study	Uses Player Data?	Method	Accuracy
Rodrigues & Dias	No	Logistic Regression	—
Atta Mills et al.	No	Random Forest, NN	Improved vs baseline
Zeileis & Hornik	Partial	Random Forest	—
Saha et al.	No	Deep Learning	63.3%
Wong et al.	Yes	ML Classifiers	—
2023 WWC Study	Yes	XGBoost	~58%
Frontiers (2025)	Yes	MLP Neural Net	86.7%
HIGFormer (2025)	Yes	Graph Transformer	Club-level only
Main Paper (2025)	Yes	5-Model Ensemble	59.38%

Rodrigues & Dias — They judged their model using betting profit margins, not accuracy. So there's no simple "% correct" number — success was measured in money made, not match predictions.

Zeileis & Hornik — Their model gives a probability for every possible match in the tournament (like a "multiverse" of outcomes), instead of one single win/loss guess. Since it's probabilistic, there's no single accuracy score to report.

Wong et al. — Their paper focused on showing that player stats help more than aggregate models², but a clean overall accuracy number wasn't clearly given in the summary — it's still a valid finding, just not a hard percentage.

Atta Mills et al. — Similar case: their key result is that Random Forest/NN did better than the traditional baseline, but not a single specific accuracy number

References

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Logistic Regression — A basic statistical model that predicts one of two outcomes (win/lose) using a weighted formula. Simple, fast, easy to interpret, but can't capture complex patterns.

Random Forest (RF) — Builds many decision trees on random subsets of data and averages their votes. Handles messy, non-linear data well — a common baseline in sports prediction.

Neural Network (NN) — A model loosely inspired by the brain, made of layered "neurons" that learn patterns through training. Good at finding complex relationships but needs more data.

Deep Learning (DNN/ANN) — A neural network with many layers ("deep"), able to learn very complex, high-level patterns. Used when there's enough data to justify the extra complexity.

ML Classifiers (general term) — Just means "a machine learning model used to sort data into categories" (like Win/Loss). Could be any of the above — the term is used when a study doesn't commit to one specific algorithm.

XGBoost — A powerful "boosting" method that builds trees one after another, each one fixing the mistakes of the last. Known for high accuracy on structured/tabular data like match stats.

MLP (Multilayer Perceptron) — A specific, simple type of neural network with fully connected layers. It's the most common "basic" deep learning model — good middle ground between simple ML and heavy deep learning.

Graph Transformer — Combines graphs (players as connected dots) with a Transformer (the architecture behind models like ChatGPT) to understand how players interact with each other, not just their individual stats.

5-Model Ensemble (used in the main paper) — Instead of picking one method, it runs 5 models (Logistic Regression, Random Forest, XGBoost, AdaBoost, KNN) together and takes a majority vote. The idea: combining different models' strengths usually beats any single model alone.

(Bonus, since it's in the ensemble): **AdaBoost** — Trains models one after another, giving extra focus to the mistakes each previous model made. **KNN (K-Nearest Neighbors)** — Predicts an outcome by looking at the most ↓ lar past matches and going with what happened most often in those cases.